# EC-333 Microprocessor and Interfacing Techniques Experiment # 13

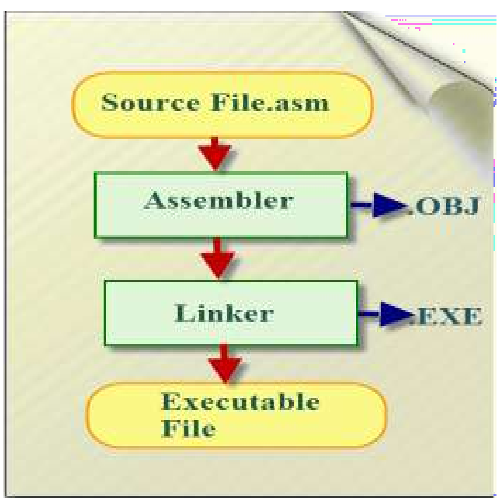
## 1. Objective/s:

## 2. Theory

The assembly language is a low-level programming language used to write program code in terms of mnemonics. Even though there are many high-level languages that are currently in demand, assembly programming language is popularly used in many applications. It can be used for direct hardware manipulations. It is also used to write the 8051 programming code efficiently with less number of clock cycles by consuming less memory compared to the other high-level languages.

### 2.1. 8051 Programming in Assembly Language

The assembly language is a fully hardware related programming language. The embedded designers must have sufficient knowledge on hardware of particular processor or controllers before writing the program. The assembly language is developed by mnemonics; therefore, users cannot understand it easily to modify the program.

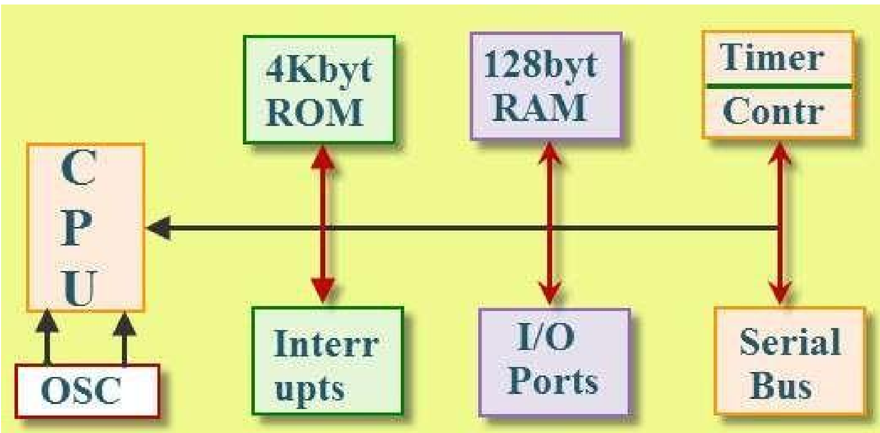



8051 Programming in Assembly Language

Assembly programming language is developed by various compilers and the "keiluvison" is best suitable for microcontroller programming development. Microcontrollers or processors can understand only binary language in the form of '0s or 1s'; An assembler converts the assembly language to binary language, and then stores it in the microcontroller memory to perform the specific task.

### 2.1. 8051 Microcontroller Architecuture

The 8051 microcontroller is the CISC based Harvard architecture, and it has peripherals like 32 I/O, timers/counters, serial communication and memories. The microcontroller requires a program to perform the operations that require a memory for saving and to read the functions. The 8051 microcontroller consists of RAM and ROM memories to store instructions.




**8051 Microcontroller Architecture**

A Register is the main part in the processors and microcontrollers which is contained in the memory that provides a faster way of collecting and storing the data. The 8051 assembly language programming is based on the memory registers. If we want to manipulate data to a processor or controller by performing subtraction, addition, etc., we cannot do that directly in the memory, but it needs registers to process and to store the data. Microcontrollers contain several types of registers that can be classified according to their instructions or content that operate in them.

**Rules of Assembly Language**

- The assembly code must be written in upper case letters
- The labels must be followed by a colon (label :)
- All symbols and labels must begin with a letter
- All comments are typed in lower case
- The last line of the program must be the END directive
  - The assembly language mnemonics are in the form of op-code, such as MOV, ADD, JMP, and so on, which are used to perform the operations.

- **Op-code:** The op-code is a single instruction that can be executed by the CPU. Here the op-code is a MOV instruction.
- **Operands:** The operands are a single piece of data that can be operated by the op-code. Example, multiplication operation is performed by the operands that are multiplied by the operand.

## 3. Lab Task

1. Write a program to calculate the sum of 3 number.
2. Write a program that subtract one number from the other.